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Food Delivery System

Introduction:

A Food Delivery System allows customers to order food from restaurants through a website or mobile application. Customers can browse menus, place orders, make payments, track delivers, and rate their experience.

1. Candidate Conceptual Classes

Conceptual classes are the important real world concepts represented in a system. For a Food Delivery System, candidate classes can be identified using the noun phrase technique and category list technique.

A. Noun-Phrase Technique

The following nouns from the system description can be treated as candidate conceptual.

- Restaurant
- Menu
  Food Item
  Order
  Customer
  Delivery driver
  Rating
  Payment
  Delivery Address
- Notification

B. Category-List Technique

Classes can also be identified by grouping the system's concepts into categories:

- People: Customer, Delivery Driver, Restaurant Staff
- Places and Businesses: Restaurant, Delivery Address

  Products: Menu, Food Item
- Transactions: Order, Payment
- Feedback and Communication: Rating, Notification

2. Attributes and Methods of Three Important Classes

Attributes represent the state or data of an object, while methods represent its behavior or actions.

Class: Restaurant

Attributes (state)	Methods (behavior)
restaurantID	displayMenu()
name	updateMenu()
address	acceptOrder()
phoneNumber	updateOpeningHours()
openingHours	viewOrders()

Class: Order

Attributes (state)	Methods (behavior)
orderID	createObject()
orderDate	addItem()
foodItems	calculateTotal()
totalAmount	cancelOrder()
orderStatus	trackOrder()

Class: Delivery Driver

Attributes (state)	Methods (behavior)
driverID	acceptDelivery()
name	pickUpOrder()
phoneNumber	updateLocation()
vehicleNumber	deliverOrder()
avaliabilityStatus	uodateAvailability()

Note: The attributes and methods can very according to the system's requirements. For Example, a delivery service may also include live GPS tracking and online Payment.

3. Abstraction Decisions

Abstraction means showing only the important details of an object and hiding unnecessary details. In the Food Delivery System, I included the restaurant name, menu, food prices, order status, and delivery driver's contact details because they are needed to order and deliver food. I ignored unrelated details such as a customer's eye color, clothing, or hobbies because they do not affect the delivery process. I also ignore the internal mechanical details of the driver's vehicle because the system mainly needs the driver's identity, availability, and delivery status.

4. One Invented Software Object

Class Name: Delivery Tracker

Delivery Tracker is an invented software object that does not exist as a physical object in the real world. It helps customers and the system track the progress of a food delivery. It can receive location updates from the delivery driver, show the current delivery status, and estimate the arrival time. This object makes delivery information easier to access and helps customers know when their food is excepted to arrive.

Conclusion

The Food Delivery System uses conceptual classes to represent restaurants, menus, orders, delivery drivers, and ratings. Attributes describe the data of each object, while methods describe the actions it can perform. Abstraction keeps the system focused on relevant information, and the invented Delivery Tracker object supports delivery tracking.